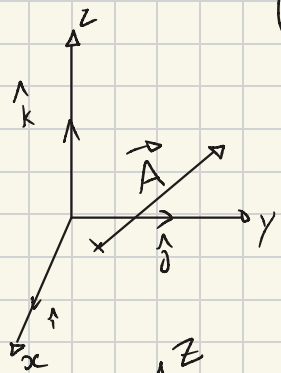


Lecture 1: Dot product

Vectors: { * Direction
* magnitude / length



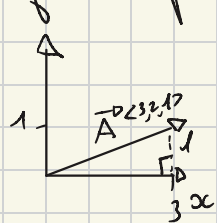
$$\vec{A} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k} = \langle a_1, a_2, a_3 \rangle$$

Length: $|\vec{A}|$ (a scalar); direction: $\text{dir}(\vec{A})$

$$\vec{A} = 3\hat{i} + 2\hat{j} + \hat{k} = \langle 3, 2, 1 \rangle \quad |\vec{A}|?$$

Imagine vector $\vec{B} = \langle 3, 2 \rangle$ so $\vec{A}$ is $\vec{B}$ with $1\hat{k}$ on top of it

If we represent it in 2D:

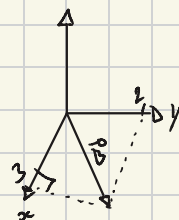


$$\text{Pythagore: } |\vec{A}|^2 = |\vec{B}|^2 + 1^2$$

$$\Leftrightarrow |\vec{A}| = \sqrt{|\vec{B}|^2 + 1^2}$$

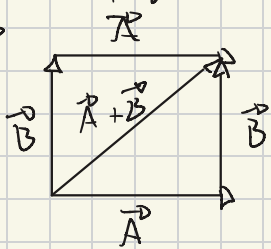
$$\Leftrightarrow |\vec{A}| = \sqrt{3^2 + 1^2} \Leftrightarrow \sqrt{10} \Leftrightarrow \sqrt{10}$$

$$|\vec{B}|^2 = 3^2 + 2^2 = 13$$



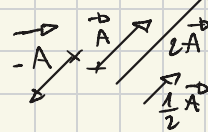
In general: $\vec{A} = \langle a_1, a_2, a_3 \rangle: |\vec{A}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

Addition: $\vec{A} + \vec{B}$



$$\vec{A} + \vec{B} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

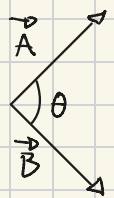
Multiplication by scalars:



Dot Product:

Def: $\vec{A} \cdot \vec{B} = \sum a_i b_i = \overbrace{a_1 b_1 + a_2 b_2 + a_3 b_3}^{\text{is a scalar}}$

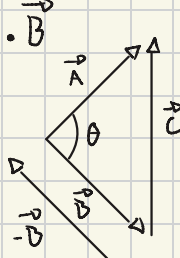
Geometr. ally: $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos(\theta)$



What does geom definition mean?

1) $\vec{A} \cdot \vec{A} = |\vec{A}|^2 \underbrace{\cos(0)}_{=1} = a_1^2 + a_2^2 + a_3^2$

2) $\vec{A} \cdot \vec{B}$



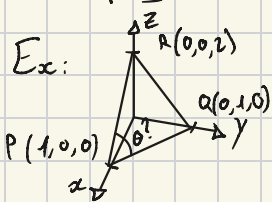
$\vec{C} = \vec{A} - \vec{B}$

Law of cosines: $|\vec{C}|^2 = |\vec{A}|^2 + |\vec{B}|^2 - 2|\vec{A}||\vec{B}|\cos\theta$

$$|\vec{C}|^2 = \vec{C} \cdot \vec{C} = (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B}) = \underbrace{\vec{A} \cdot \vec{A}}_{|\vec{A}|^2} - \vec{A} \cdot \vec{B} - \vec{B} \cdot \vec{A} + \underbrace{\vec{B} \cdot \vec{B}}_{|\vec{B}|^2} = |\vec{A}|^2 + |\vec{B}|^2 - 2\vec{A} \cdot \vec{B}$$

Applications:

1) Computing lengths and angles



$$\vec{PQ} \cdot \vec{PR} = |\vec{PQ}| |\vec{PR}| \cos(\theta) \Leftrightarrow \frac{\vec{PQ} \cdot \vec{PR}}{|\vec{PQ}| |\vec{PR}|} = \cos \theta$$

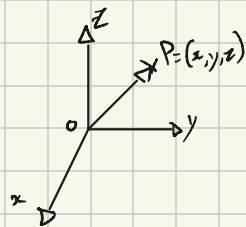
sign of $\vec{A} \cdot \vec{B}$: > 0 if $\theta < 90^\circ$
 $= 0$ if $\theta = 90^\circ$
 < 0 if $\theta > 90^\circ$

$$\frac{\vec{PQ} \cdot \vec{PR}}{|\vec{PQ}| |\vec{PR}|} = \frac{\langle -1, 1, 0 \rangle \cdot \langle -1, 0, 2 \rangle}{\sqrt{(-1)^2 + 1^2 + 0^2} \sqrt{(-1)^2 + 0^2 + 2^2}} = \frac{1+0+0}{\sqrt{2} \sqrt{5}} = \frac{1}{\sqrt{10}} = \cos \theta \quad \theta = \cos^{-1}\left(\frac{1}{\sqrt{10}}\right) \approx 71.5^\circ$$

2) Detection of orthogonality: Ex: $x + 2y + 3z = 0$ is eq. of a plane

Get: plane through O, perpendicular to $\vec{A}$

$$\vec{A} \cdot \vec{B} = 0 \text{ if } \cos \theta = 0 \Leftrightarrow \theta = 90^\circ \Leftrightarrow \vec{A} \perp \vec{B}$$



$$\vec{OP} = \langle x, y, z \rangle$$

$$\vec{A} = \langle 1, 2, 3 \rangle$$

$$\vec{OP} \cdot \vec{A} = 0 \Leftrightarrow \vec{OP} \perp \vec{A} \text{ (perpendicular)}$$